Cse 421- Computer Networks

Lab Project

Topic -07 ; Marine World

Sec- 01

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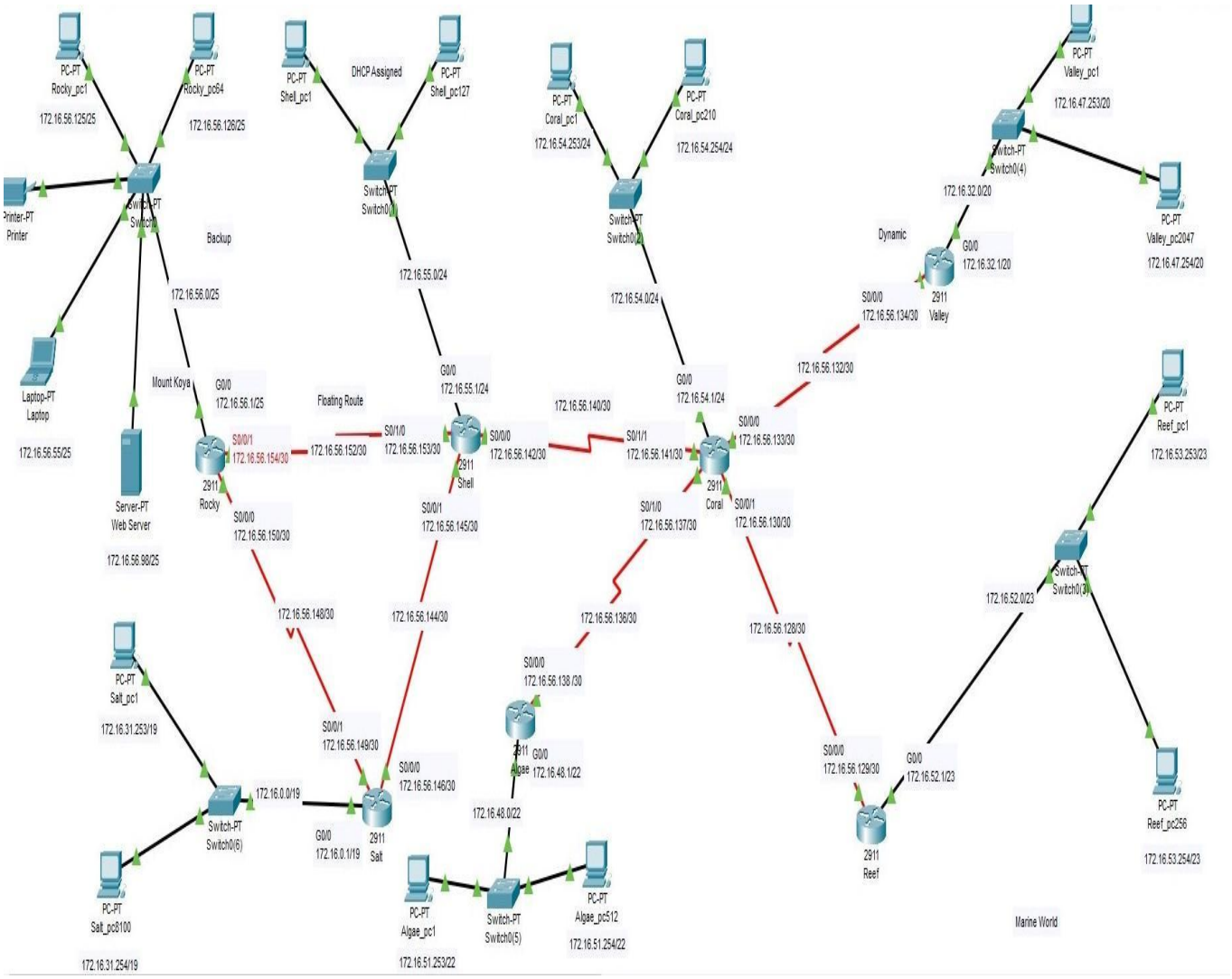
Report On Marine World Networking

Network Topology Diagram:



VLSM Table: Network address --- 172.16.1.0/24

Name	Hosts Needed	Network Address	Mask	Usable Range
Salt	8100	172.16.0.0/19	255.255.224.0	172.16.0.1 - 172.16.31.254
Valley	2047	172.16.32.0/20	255.255.240.0	172.16.32.1 - 172.16.47.254
Algae	512	172.16.48.0/22	255.255.252.0	172.16.48.1 - 172.16.51.254
Reef	256	172.16.52.0/23	255.255.254.0	172.16.52.1 - 172.16.53.254
Coral	210	172.16.54.0/24	255.255.255.0	172.16.54.1 - 172.16.54.254
Shell	127	172.16.55.0/24	255.255.255.0	172.16.55.1 - 172.16.55.254
Rocky	64	172.16.56.0/25	255.255.255.128	172.16.56.1 - 172.16.56.126
CR	2	172.16.56.128/30	255.255.255.252	172.16.56.129 - 172.16.56.130
CV	2	172.16.56.132/30	255.255.255.252	172.16.56.133 - 172.16.56.134
CA	2	172.16.56.136/30	255.255.255.252	172.16.56.137 - 172.16.56.138
CSh	2	172.16.56.140/30	255.255.255.252	172.16.56.141 - 172.16.56.142
ShS	2	172.16.56.144/30	255.255.255.252	172.16.56.145 - 172.16.56.146
SRo	2	172.16.56.148/30	255.255.255.252	172.16.56.149 - 172.16.56.150
ShRo	2	172.16.56.152/30	255.255.255.252	172.16.56.153 - 172.16.56.154

IP TABLE DEVICE:

Device	IP address	Subnet mask	Default gateway
Coral_PC1	172.16.54.253/24	255.255.255.0	172.16.54.1
Coral_PC210	172.16.54.254/24	255.255.255.0	172.16.54.1
Valley_PC1	172.16.47.253/20	255.255.240.0	172.16.32.1
Valley_PC247	172.16.47.254/20	255.255.240.0	172.16.32.1
Reef_PC1	172.16.53.253/23	255.255.254.0	172.16.52.1
Reef_PC256	172.16.53.254/23	255.255.254.0	172.16.52.1
Algae_PC1	172.16.51.253/22	255.255.252.0	172.16.48.1
Algae_PC512	172.16.51.254/22	255.255.252.0	172.16.48.1
Shell_PC1	**DHCP assigned**	**DHCP assigned**	N/A
Shell_PC127	**DHCP assigned**	**DHCP assigned**	N/A
Salt_PC1	172.16.31.253/19	255.255.224.0	172.16.0.1
Salt_PC8100	172.16.31.254/19	255.255.224.0	172.16.0.1
Rockey_PC1	172.16.56.125/25	255.255.255.128	172.16.56.1
Rockey_PC64	172.16.56.126/25	255.255.255.128	172.16.56.1
Rockey_Laptop	172.16.56.55/25	255.255.255.128	172.16.56.1
Rockey_Printer	N/A	N/A	172.16.56.1
Rockey_WebServer	172.16.56.98/25	255.255.255.128	172.16.56.1

Routing Information table:

Device	Interface	IP address	Subnet mask
Coral	G0/0	172.16.54.1/24	255.255.255.0
	S0/0/0	172.16.56.133/30	255.255.255.252
	S0/0/1	172.16.56.130/30	255.255.255.252
	S0/1/0	172.16.56.137/30	255.255.255.252
	S0/1/1	172.16.56.141/30	255.255.255.252
Valley	G0/0	172.16.32.1/20	255.255.240.0
	S0/0/0	172.16.56.134/30	255.255.255.252
Reef	G0/0	172.16.52.1/23	255.255.254.0
	S0/0/0	172.16.56.129/30	255.255.255.252
Algae	G0/0	172.16.48.1/22	255.255.252.0
	S0/0/0	172.16.56.138 /30	255.255.255.252
Shell	G0/0	172.16.55.1/24	255.255.255.0
	S0/0/0	172.16.56.142/30	255.255.255.252
	S0/0/1	172.16.56.145/30	255.255.255.252
	S0/1/0	172.16.56.153/30	255.255.255.252
Salt	G0/0	172.16.0.1/19	255.255.224.0
	S0/0/0	172.16.56.146/30	255.255.255.252
	S0/0/1	172.16.56.149/30	255.255.255.252
Rocky	G0/0	172.16.56.1/25	255.255.255.128
	S0/0/0	172.16.56.150/30	255.255.255.252
	S0/0/1	172.16.56.154/30	255.255.255.252

Configuration:

PC ip address assigning :

Reef_pc1 -> Desktop -> ip configuration
Ipv4 Address-172.16.53.253
Subnet mask-255.255.254.0
Default gateway-172.16.52.1 //255.255.254.0
Reef_pc256 -> Desktop -> ip configuration
Ipv4 Address-172.16.53.254
Subnet mask-255.255.254.0
Default gateway-172.16.52.1 // 255.255.254.0

Coral_pc1 -> Desktop -> ip configuration
Ipv4 Address-172.16.54.253
Subnet mask-255.255.255.0
Default gateway-172.16.54.1 //255.255.255.0
Coral_pc210 -> Desktop -> ip configuration
Ipv4 Address-172.16.54.254
Subnet mask-255.255.255.0
Default gateway-172.16.54.1 //255.255.255.0

Valley_pc1 -> Desktop -> ip configuration
Ipv4 Address-172.16.47.253
Subnet mask-255.255.240.0
Default gateway- 172.16.32.1 //255.255.240.0
Valley_pc2047 -> Desktop -> ip configuration
Ipv4 Address-172.16.47.254
Subnet mask-255.255.240.0
Default gateway-172.16.32.1 //255.255.240.0

Algae_pc1 -> Desktop -> ip configuration
Ipv4 Address-172.16.51.253
Subnet mask-255.255.252.0
Default gateway-172.16.48.1 // 255.255.252.0
Algae_pc512 -> Desktop -> ip configuration
Ipv4 Address-172.16.51.254

Subnet mask-255.255.252.0
Default gateway-172.16.48.1 //255.255.252.0

Shell_pc1 -> Desktop -> ip configuration
Ipv4 Address-172.16.55.253
Subnet mask-255.255.255.0
Default gateway-172.16.55.1 //255.255.255.0
Shell_pc127 -> Desktop -> ip configuration
Ipv4 Address-172.16.55.254
Subnet mask-255.255.255.0
Default gateway-172.16.55.1 //255.255.255.0

Salt_pc1 -> Desktop -> ip configuration
Ipv4 Address-172.16.31.253
Subnet mask-255.255.224.0
Default gateway-172.16.0.1 //255.255.224.0
Salt_pc8100 -> Desktop -> ip configuration
Ipv4 Address-172.16.31.254
Subnet mask-255.255.224.0
Default gateway-172.16.0.1 //255.255.224.0

Rocky_pc1 -> Desktop -> ip configuration
Ipv4 Address-172.16.56.125
Subnet mask-255.255.255.128
Default gateway-172.16.56.1 //255.255.255.128
Rocky_pc64 -> Desktop -> ip configuration
Ipv4 Address-172.16.56.126
Subnet mask-255.255.255.128
Default gateway-172.16.56.1 //255.255.255.128
Laptop -> Desktop -> ip configuration
Ipv4 Address-172.16.56.55
Subnet mask-255.255.255.128
Default gateway-172.16.56.1 //255.255.255.128
Printer -> Desktop -> ip configuration
Default gateway-172.16.56.1 //255.255.255.128
Web Server -> Desktop -> ip configuration
Ipv4 Address-172.16.56.98
Subnet mask-255.255.255.128
Default gateway-172.16.56.1 //255.255.255.128

Router Ip Address Assigning:

Router : Reef -> CLI	Router : Valley-> CLI
En Conf t Interface g0/0 Ip address 172.16.52.1 255.255.254.0 No shutdown Exit Interface s 0/0/0 Ip address 172.16.56.129 255.255.255.252 No shutdown Exit	En Conf t Interface g0/0 Ip address 172.16.32.1 255.255.240.0 No shutdown Exit Interface s 0/0/0 Ip address 172.16.56.134 255.255.255.252 No shutdown Exit

Router : Salt>CLI	Router : Valley-> CLI
En Conf t Interface g0/0 Ip address 172.16.0.1 255.255.224.0 No shutdown Exit Interface s0/0/0 Ip address 172.16.56.146 255.255.255.252 No shutdown Exit Interface s0/0/1 Ip address 172.16.56.149 255.255.255.252 No shutdown Exit	En Conf t Interface g0/0 Ip address 172.16.32.1 255.255.240.0 No shutdown Exit Interface s 0/0/0 Ip address 172.16.56.134 255.255.255.252 No shutdown Exit

Router : Coral -> CLI

```
En
Conf t
Interface g0/0
Ip address  172.16.54.1  255.255.255.0
No shutdown
Exit
Interface   s 0/0/0
Ip address    172.16.56.133    255.255.255.252
No shutdown
Exit
Interface  s 0/0/1
Ip address   172.16.56.130     255.255.255.252
No shutdown
Exit
Interface   s 0/1/0
Ip address   172.16.56.137    255.255.255.252
No shutdown
Exit
Interface   s 0/1/1
Ip address  172.16.56.141    255.255.255.252
No shutdown
Exit
```

Router : Algae->CLI

```
En
Conf t
Interface g0/0
Ip address  172.16.48.1  255.255.252.0
No shutdown
Exit
Interface   s 0/0/0
Ip address     172.16.56.138    255.255.255.252
No shutdown
Exit
```

Router : Shell>CLI

```
En
Conf t
Interface g0/0
Ip address  172.16.55.1  255.255.255.0
No shutdown
Exit
Interface  s0/0/0
Ip address  172.16.56.142   255.255.255.252
No shutdown
Exit
Interface  s0/0/1
Ip address 172.16.56.145   255.255.255.252
No shutdown
Exit
Interface  s0/1/0
Ip address 172.16.56.153  255.255.255.252
No shutdown
Exit
```

Dynamic Routing:

Router Valley -> CLI	Router Coral -> CLI
En	En
conf t	conf t
ip route 0.0.0.0 0.0.0.0 s0/0/0	ip route 0.0.0.0 0.0.0.0 s0/0/0
router rip	router rip
version 2	version 2
no auto-summary	no auto-summary
network 172.16.56.132 (network address of CV)	network 172.16.56.132 (network address of CV)
network 172.16.32.0 (network address of Valley)	network 172.16.54.0 (network address of Coral)
default-information originate	default-information originate
End	End
copy run start	copy run start
Enter	Enter

Floating route in Shell:

Router Salt -> CLI

En

Conf t

Ip route 172.16.56.152 255.255.255.252 s0/1/0 5 (network address of shro shro mask exit interface from shell distance)

Exit

Static routing in Coral:

Router Coral -> CLI

```
en
conf t
Ip route 172.16.52.0  255.255.254.0   172.16.56.129          // to reef  s0/0/0 (Reef)
Ip route  172.16.48.0  255.255.252.0  172.16.56.138           // algae  s0/0/0 (Algae)
Ip route  172.16.55.0  255.255.255.0  172.16.56.142          //to shell   s0/0/0 (Shell)
Ip route  172.16.0.0  255.255.224.0    172.16.56.142           //to salt  s0/0/0 (Shell)
Ip route  172.16.56.0  255.255.255.128    172.16.56.142       //to rocky s0/0/0 (Shell)
exit
```

Router Algae -> CLI

```
en
conf t
Ip route 172.16.54.0 255.255.255.0  172.16.56.137              // to coral s0/1/0 (Coral)
Ip route 172.16.52.0 255.255.254.0  172.16.56.137              // to reef  s0/1/0 (Coral)
Ip route 172.16.32.0  255.255.240.0  172.16.56.137            // to valley  s0/1/0 (Coral)
Ip route 172.16.55.0  255.255.255.0 172.16.56.137              // to shell  s0/1/0 (Coral)
Ip route  172.16.0.0  255.255.224.0  172.16.56.137              // to salt  s0/1/0 (Coral)
Ip route  172.16.56.0   255.255.255.128   172.16.56.137       // to rocky  s0/1/0 (Coral)
Exit
```

Router Reef -> CLI

```
en
conf t
Ip route 172.16.54.0   255.255.255.0   172.16.56.130          // to coral  s0/0/1(Coral)
Ip route  172.16.48.0  255.255.252.0  172.16.56.130           // to algae   s0/0/1(Coral)
Ip route 172.16.32.0  255.255.240.0 172.16.56.130             // to valley   s0/0/1(Coral)
Ip route 172.16.55.0  255.255.255.0 172.16.56.130             // to shell   s0/0/1(Coral)
Ip route  172.16.0.0  255.255.224.0  172.16.56.130             // to salt   s0/0/1(Coral)
Ip route  172.16.56.0 255.255.255.128  172.16.56.130           //to rocky   s0/0/1(Coral)
exit
```

Router Shell-> CLI

```
-en
-conf t
Ip route 172.16.54.0 255.255.255.0 172.16.56.141              // to coral s0/1/1(Coral)
Ip route  172.16.48.0  255.255.252.0  172.16.56.141           // to algae s0/1/1(Coral)
Ip route 172.16.32.0  255.255.240.0 172.16.56.141             // to valley  s0/1/1(Coral)
Ip route 172.16.52.0 255.255.254.0 172.16.56.141              // to reef  s0/1/1(Coral)
Ip route  172.16.0.0  255.255.224.0  172.16.56.146             // to salt  s0/0/0(Salt)
Ip route  172.16.56.0 255.255.255.128  172.16.56.154          / to rocky  s0/0/1(Rocky)
exit
```

Router Salt-> CLI

-en

-conf t

Ip route 172.16.54.0 255.255.255.0 172.16.56.145 // to coral s0/0/1(Shell)

Ip route 172.16.48.0 255.255.252.0 172.16.56.145 // to algae s0/0/1(Shell)

Ip route 172.16.32.0 255.255.240.0 172.16.56.145 // to valley s0/0/1(Shell)

Ip route 172.16.52.0 255.255.254.0 172.16.56.145 // to reef s0/0/1(Shell)

Ip route 172.16.55.0 255.255.255.0 172.16.56.145 // to Shell s0/0/1(Shell)

Ip route 172.16.56.0 255.255.255.128 172.16.56.150 // to rocky s0/0/0(Rocky)

exit

Router Rocky-> CLI

-en

-conf t

Ip route 172.16.54.0 255.255.255.0 172.16.56.153 //to coral s0/1/0(Shell)

Ip route 172.16.48.0 255.255.252.0 172.16.56.153 // to algae s0/1/0(Shell)

Ip route 172.16.32.0 255.255.240.0 172.16.56.153 // to valley s0/1/0(Shell)

Ip route 172.16.52.0 255.255.254.0 172.16.56.153 //to reef s0/1/0(Shell)

Ip route 172.16.55.0 255.255.255.0 172.16.56.153 // to Shell s0/1/0(Shell)

Ip route 172.16.0.0 255.255.224.0 172.16.56.149 //to salt s0/0/1(Salt)

Exit

Backup Routing:

Router Valley -> CLI En conf t ip route 0.0.0.0 0.0.0.0 s0/0/0(exit interface of valley) Exit	**Router Reef -> CLI** En conf t ip route 0.0.0.0 0.0.0.0 s0/0/0(exit interface of Reef) Exit
Router Algae -> CLI En conf t ip route 0.0.0.0 0.0.0.0 s0/0/0(exit interface of Algae) Exit	**Router Coral-> CLI** En conf t ip route 0.0.0.0 0.0.0.0 s0/1/1(exit interface of Coral) Exit
Router Shell-> CLI En conf t ip route 0.0.0.0 0.0.0.0 s0/1/0(exit interface of Shell) Exit	**Router Salt-> CLI** En conf t ip route 0.0.0.0 0.0.0.0 s0/0/1(exit interface of Salt) Exit
Router Rocky> CLI En conf t ip route 0.0.0.0 0.0.0.0 g0/0(exit interface of Rocky) Exit	

DHCP configuration for shell

Router Shell -> CLI

en
conf t
ip dhcp pool Shell-LAN
network 172.16.55.0 255.255.255.0 (Shell network address shell subnet mask)
default-router 172.16.55.1 (Shell gigabyte interface ip address)
Exit

Shell_PC1->Config-> select DHCP
Shell_PC127->Config-> select DHCP

Testing Commands:

(The commands should be in enable mode)

- Ping (ip address)
- Show ip route
- Ipconfig /all //command for in pc
- Show run
- Show ip dhcp binding //To display the IP address bindings and their associated leases
- Show ip dhcp conflict //any IP address conflicts that the router has detected in the DHCP address pool
- Show Ip interface brief